

Subject: Azure Fundamentals

Unit 1 : Cloud Concepts

Computer Science & Engineering

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Outline

- **Introduction of Cloud Computing**
- **Cloud computing principles**
- **Azure Fundamentals, History, Characteristics**

Azure Fundamentals

Btech, 5th Semester

Prerequisite: Basic understanding of computer concepts and basic programming

Rationale: This course provides a broad introduction to Azure cloud , infrastructure , services, security and compliance ,also billing , pricing and support plans.

Introduction of Cloud Computing

Definition: Cloud computing refers to the on demand provision of computational resources (data, software) via a computer network, rather than from a local computer.



Cloud Computing



What is Cloud Computing

1. Cloud computing is the delivery of computing services over the internet. Computing services include common IT infrastructure such as virtual machines, storage, databases, and networking.
2. Cloud services also expand the traditional IT offerings to include things like Internet of Things (IoT), machine learning (ML), and artificial intelligence (AI).
3. Because cloud computing uses the internet to deliver these services, it doesn't have to be constrained by physical infrastructure the same way that a traditional datacenter is.
4. That means if you need to increase your IT infrastructure rapidly, you don't have to wait to build a new datacenter—you can use the cloud to rapidly expand your IT footprint.



Introduction of Cloud Computing

- Microsoft Azure is a cloud computing platform with an ever-expanding set of services to help you build solutions to meet your business goals.
- Azure services support everything from simple to complex. Azure has simple web services for hosting your business presence in the cloud.
- Azure also supports running fully virtualized computers managing your custom software solutions.
- Azure provides a wealth of cloud-based services like remote storage, database hosting, and centralized account management.
- Azure also offers new capabilities like artificial intelligence (AI) and Internet of Things (IoT) focused services.

What is Azure Fundamentals?

- Azure Fundamentals is a series of three learning paths that familiarize you with Azure and its many services and features.
- Whether you're interested in compute, **networking, or storage services**; learning about cloud security best practices; or exploring governance and management options, think of Azure Fundamentals as your curated guide to Azure.
- Azure Fundamentals includes interactive exercises that give **you hands-on experience** with Azure. Many exercises **provide a temporary Azure portal environment** called the sandbox, which allows you to practice creating cloud resources for free at your own pace.
- Technical IT experience isn't required; however, having general IT knowledge will help you get the most from your learning experience.



HISTORY OF CLOUD COMPUTING

- Before emerging the cloud computing, there was Client/Server computing which is basically a centralized storage in which all the software applications, all the data and all the controls are resided on the server side.
- If a single user wants to access specific data or run a program, he/she need to connect to the server and then gain appropriate access, and then he/she can do his/her business.
- Then after, distributed computing came into picture, where all the computers are networked together and share their resources when needed.
- On the basis of above computing, there was emerged of cloud computing concepts that later implemented.
- At around in 1961, John MacCharty suggested in a speech at MIT that computing can be sold like a utility, just like a water or electricity. It was a brilliant idea, but like all brilliant ideas, it was ahead if its time, as for the next few decades, despite interest in the model, the technology simply was not ready for it.



HISTORY OF CLOUD COMPUTING

- But of course time has passed and the technology caught that idea and after few years we mentioned that:
- In 1999, Salesforce.com started delivering of applications to users using a simple website. The applications were delivered to enterprises over the Internet, and this way the dream of computing sold as utility were true.
- In 2002, Amazon started Amazon Web Services, providing services like storage, computation and even human intelligence. However, only starting with the launch of the Elastic Compute Cloud in 2006 a truly commercial service open to everybody existed.
- In 2009, Google Apps also started to provide cloud computing enterprise applications.
- Of course, all the big players are present in the cloud computing evolution, some were earlier, some were later. In 2009, Microsoft launched Windows Azure, and companies like Oracle and HP have all joined the game. This proves that today, cloud computing has become mainstream.

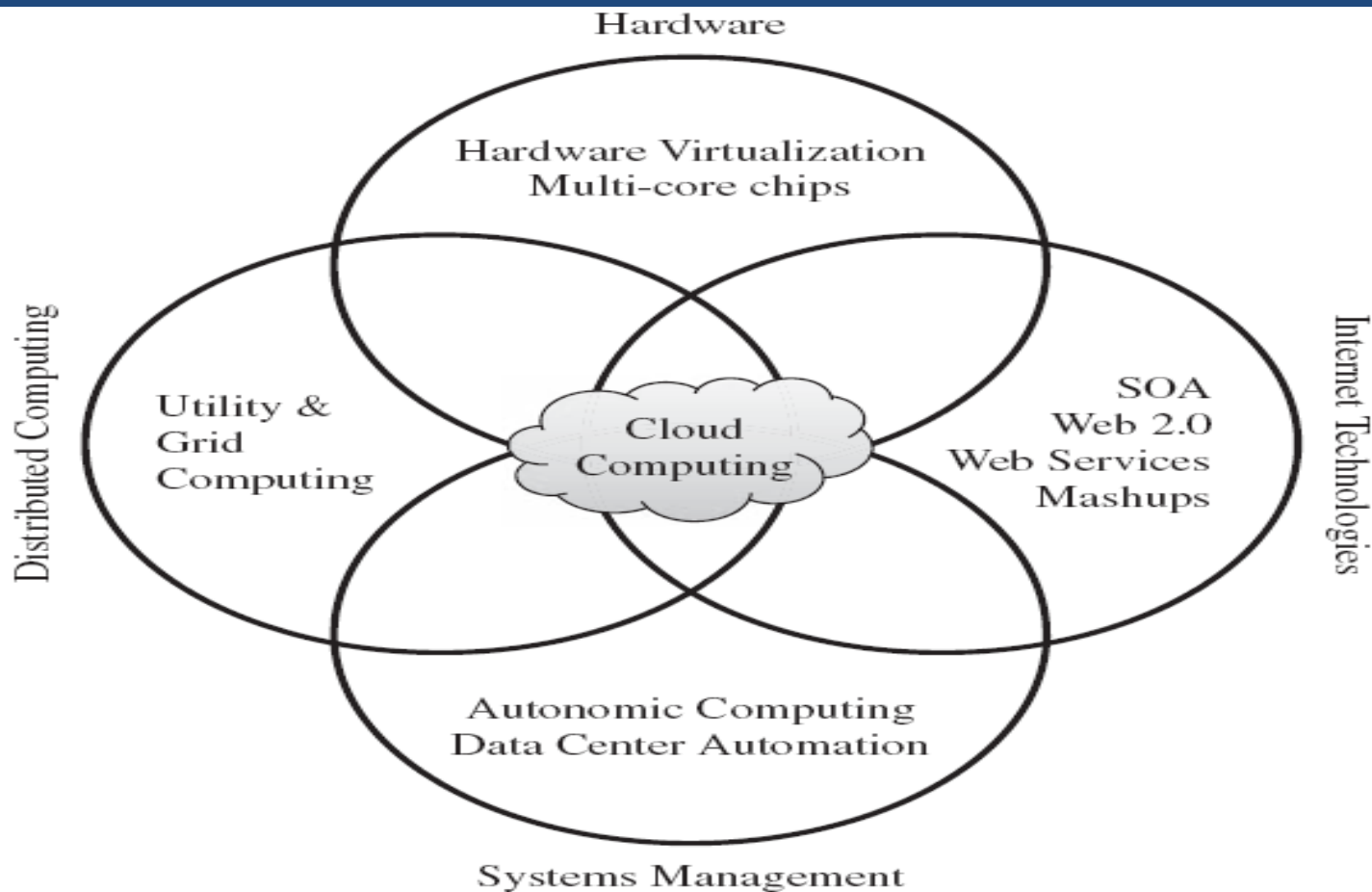
Common Characteristic

Cloud should have

- (i) pay-per-use (no ongoing commitment, utility prices)
- (ii) elastic capacity and the illusion of infinite resources
- (iii) self-service interface
- (iv) virtualized (abstracted) resources

The ultimate goal is allowing customers to run their everyday IT infrastructure “in the cloud.”

Roots of Cloud Computing



Internet Technologies

1. Web Service

- applications running on different messaging product platforms
- enabling information from one application to be made available to others
- enabling internal applications to be made available over the Internet

2. SOA

- address requirements of loosely coupled, standards-based, and protocol-independent distributed computing
- WS ,HTTP, XML
 - Common mechanism for delivering service
- applications is a collection of services that together perform complex business logic
- Building block in IaaS
 - User authentication, payroll management, calender

Distributed Computing

1. Grid Computing

- aggregation of distributed resources
- transparently access
- Problems
 - QoS, Lack of performance
 - Availability, Virtualization

2. Utility Computing

- assign a “utility” value to users
- QoS constraints (deadline, importance, satisfaction)

Hardware

- Hardware Virtualization
 - overcome most operational issues of data center building and maintenance
 - improving sharing and utilization of computer systems
 - running multiple operating systems and software stacks on a single physical platform
 - virtual machine monitor (VMM)
 - mediates access to the physical hardware
 - presenting to guest operating system a virtual machine (VM)
 - is a set of virtual platform interfaces

Thank You!!!

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